Siim Hirse

Senior Software Engineer

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EDUCATION

Master's Degree in Computer Science
Tallinn University of Technology
Tallinn, Estonia
2011 - 2016

COMPETENCIES

- Cloud-Native Architecture & Distributed Systems
- High-Performance API Design & Backend Engineering
- Microservices & Event-Driven Systems
- AI & Machine Learning Integration
- Scalable SaaS Platform Development
- Real-Time Data Processing & Streaming Workflows
- Secure Authentication, Authorization & RBAC
- Relational & NoSQL Data Modeling
- Containerization & Kubernetes Orchestration
- CI/CD Automation & DevOps Practices
- Multi-Tenant & Multi-Region System Design
- Enterprise System Integration & API Governance
- Interactive Data Visualization & 3D Systems
- Test-Driven Development & Quality Engineering
- Agile Scrum Delivery & Cross-Functional Leadership

SUMMARY

Senior Software Engineer with 10+ years of experience delivering scalable platforms across Healthcare, Renewable Energy, Solar Tech, and FinTech industries. Specialized in backend architecture, distributed systems, real-time data processing, and AI-driven applications using modern cloud-native technologies. Proven track record of building secure microservices, high-performance APIs, interactive data visualizations, and production-grade CI/CD pipelines in AWS environments.

EXPERIENCE

Senior Software Engineer

Mar 2023 - Apr 2026

Apriorit | Poznan, Poland

- **Healthcare & Sports Analytics Industry:** Engineered a high-performance impact analytics backend using **Python**, **FastAPI**, and **AWS**, delivering 4× faster biomechanical data processing for athlete injury research.
- Designed distributed data orchestration services with **GraphQL**, **Apollo Client**, and **Redis**, unifying multi-sensor and video streams into real-time research endpoints.
- Built immersive 3D visualization dashboards using **TypeScript**, **Next.js**, and **Three.js**, transforming raw telemetry into synchronized playback, violin plots, and 3D heatmaps.
- Implemented secure research data architecture with **PostgreSQL**, **JWT**, and **RBAC**, enforcing structured schema validation and multi-tier access governance.
- Integrated AI workflows using **TensorFlow**, **OpenAI API**, and **GitHub Copilot**, enabling automated impact classification and intelligent feature acceleration.
- Strengthened platform quality via **Jest**, **React Testing Library**, and **Pytest**, ensuring cross-layer reliability across continuous Scrum releases.
- Delivered scalable cloud infrastructure using **Docker**, **Kubernetes**, and **GitHub Actions**, establishing CI/CD pipelines for white-label expansion.

Senior Full-Stack Engineer

Nov 2021 - Feb 2023

Go Wombat | Dnipro, Ukraine

- **Renewable Energy & Sustainability Platform**: Delivered a cloud-native photovoltaic SaaS platform using Python, **Django** and **React.js**, accelerating feature delivery for PV design, financial quotations, and multi-country expansion.
- Built scalable backend services with Django REST Framework, applying strong **OOP** principles; implemented secure IAM using **AWS Cognito** integrated with **Active Directory**, and developed event-driven workflows with **AWS Lambda**, **SQS**, and **Step Functions**.
- Developed dynamic frontend modules using **React.js**, **TypeScript**, **Material UI**, HTML5, and CSS3; enhanced PV system creator, user management, financial modeling UI, and led app-wide rebranding.
- Designed and optimized data architecture with PostgreSQL (relational) and **DynamoDB** (NoSQL), supporting financial computations, geographic analytics, and multi-tenant structures.
- Containerized applications with **Docker** and deployed on **AWS ECS**; configured **API Gateway** and **S3**, and implemented automated CI/CD pipelines using GitHub Actions.
- Integrated **Google Solar API** and drone data uploads to enhance high-resolution terrain analysis and geographic visualization across Europe.

Results & Achievements

- Delivered 4× faster biomechanical data processing, enabling real-time analytics and AI-driven insights for athlete research.
- Architected and scaled cloud-native SaaS platforms and microservices, supporting global expansion and high-performance distributed systems.
- Drove measurable business impact, including 30% increase in conversion rates, through optimized system design, automation, and data-driven solutions.

SKILLS

- **Programming & Backend**
  Python, FastAPI, Django, Django REST Framework, Node.js, SQLAlchemy, Celery, GraphQL, REST APIs, Microservices Architecture
- **Frontend & Visualization**
  TypeScript, React.js, Next.js, Three.js, Material UI, Tailwind CSS, Bootstrap, HTML5, CSS3, Apollo Client
- **Cloud & DevOps**
  AWS (ECS, AWS Lambda, SQS, Step Functions, API Gateway, S3), Docker, Kubernetes, Helm, GitHub Actions, CI/CD Pipelines
- **Databases & Caching**
  PostgreSQL, DynamoDB, Redis, Relational Data Modeling, NoSQL Architecture
- **Security & Identity**
  JWT, OAuth2, RBAC, OIDC, AWS Cognito, Role-Based Access Governance
- **Testing & Observability**
  Pytest, Jest, React Testing Library, Cypress, BrowserStack, Lighthouse, Prometheus, Grafana, Loki
- **AI/ML & Automation**
  TensorFlow, OpenAI API, GenAI Workflows, Impact Classification Models, AI-Assisted Development
- **Methodologies**
  Agile, Scrum, Distributed Systems Design, Event-Driven Architecture, Multi-Tenant Systems

Full-Stack Engineer

May 2019 - Oct 2021

NERDZ LAB | Tallinn, Estonia

- **Financial Technology Industry:** Consolidated legacy systems into a unified multilingual platform using **Django REST Framework**, **FastAPI**, and **React/Next.js**, aligning global operations with corporate rebranding initiatives.
- Established secure microservices governance through **OAuth2**, structured role segmentation, and enterprise authentication layers.
- Orchestrated distributed workflows using **Celery**, enabling scalable analytics and partner automation integrations.
- Optimized performance through advanced ORM patterns with **SQLAlchemy** and intelligent caching strategies, supporting geo-redirect and localized delivery logic.
- Maintained release stability via structured API validation and regression testing within Agile Scrum environments.

Software Engineer

Jun 2016 - Apr 2019

Elogic Commerce | Tallinn, Estonia

- **Digital Commerce Industry:** Built an automated roof configurator platform using **React**, **Node.js**, and **Strapi**, increasing sales conversion by 30% through digital quotation automation.
- Developed scalable API ecosystems with **Apollo GraphQL** and RESTful services, integrating geospatial validation and compliance workflows.
- Validated cross-platform performance with **Cypress**, **BrowserStack**, and **Lighthouse**, ensuring accessibility and production-grade UI stability.
- Engineered transactional data models within relational systems to optimize pricing engines and CMS-driven product delivery.
- Deployed containerized workloads using **Helm**, implementing observability pipelines with **Prometheus**, **Grafana**, and **Loki** under Agile methodology.

PROJECTS

Global Multilingual FinTech Platform Modernization

https://www.verifone.com

Led the end to end development of a multinational FinTech web platform for Verifone, rebuilding fragmented regional websites into a unified multilingual system using Python, FastAPI, Django, React, and Next.js. The solution supported secure content governance, advanced product discovery, geo based localization, and deep third party integrations, delivering a scalable cloud deployed platform on AWS that improved user flow and global brand consistency while supporting high transaction driven enterprise operations.

Sports & Healthcare Analytics (Biomechanics) Platform

https://biocorellc.com

Led the development of a scalable biomechanics analytics platform in the Sports & Healthcare industry, enabling 4× faster impact data processing for athlete safety research. Architected a microservices-based system using Python, FastAPI, GraphQL, and AWS, supporting real-time sensor ingestion, ML-driven impact classification, and advanced 3D data visualization. Designed secure, cloud-native infrastructure with containerized services, RBAC-based access control, and CI/CD automation to ensure high performance, scalability, and future white-label readiness.